RAHEELA SHABBIR

CYBERSECURITY | SECURITY RESEARCH | PYTHON DEVELOPMENT

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PROFESSIONAL PROFILE

Cybersecurity and software development professional with 3+ years of research and technical experience at the National Center for Cyber Security. Holds an M.S. in Cyber Security and the EC-Council Certified Penetration Testing Professional (CPENT) credential. Combines hands-on security assessment, network analysis and vulnerability reporting with Python backend development, secure APIs and applied AI/ML. Experienced in translating research and security requirements into practical software tools, analytical workflows and clear technical deliverables.

CORE TECHNICAL COMPETENCIES

Offensive Security Penetration testing, vulnerability assessment, OWASP Top 10, SQLi, XSS, authentication testing, Nmap, Nessus, Burp Suite, OWASP ZAP, Metasploit, Kali Linux

Network Security & DFIR Wireshark, T-Shark, Snort, packet/traffic analysis, IOC analysis, OSINT, threat hunting, MITRE ATT&CK

Software Engineering Python, Flask, FastAPI, REST APIs, MySQL, role-based access control (RBAC), input validation, Git, Linux, BeautifulSoup

AI / Data Analysis Scikit-learn, TensorFlow, Keras, NLP, spaCy, NLTK, OpenCV, EasyOCR, Pandas, NumPy, data preprocessing and model evaluation

Professional Methods Security documentation, risk-rated reporting, remediation guidance, security best practices, ISO/IEC 27001 knowledge

PROFESSIONAL EXPERIENCE

Research Assistant - Cybersecurity & AI

2023 – Present

National Center for Cyber Security (NCCS / NCFL / NASTP) | Rawalpindi, Pakistan

- Conducted vulnerability assessment and security-testing activities across Windows, Linux and web environments; assessed common misconfigurations and application weaknesses using Nmap, Nessus, Burp Suite and related tooling.
- Analyzed packet captures and network behavior using Wireshark, T-Shark and Snort, relating suspicious traffic to indicators of compromise and documenting technical findings.
- Built a secure Python/MySQL Alumni Portal with REST APIs, RBAC, input validation and parameterized queries; automated profile enrichment, deduplication, search and reporting workflows.
- Contributed to a YouTube sentiment and keyword analytics platform; developed multilingual transcript-processing features and strengthened API validation and data-handling practices.
- Designed NLP and machine-learning pipelines for sentiment analysis, topic modeling, information extraction and predictive modeling, including dataset preparation and model evaluation.
- Implemented multilingual OCR and computer vision workflows using EasyOCR, OpenCV and deep learning methods to support document digitization and forensic research.
- Prepared technical reports and research outputs, translating assessment results and experimental findings into structured recommendations and reusable workflows.

SELECTED PROJECTS & APPLIED RESEARCH

Web Application Security Assessment | Offensive Security

Performed manual and automated testing for injection, XSS, weak authentication and security misconfigurations; documented reproducible findings and remediation recommendations.

Secure Alumni Portal & Data Automation | Python / MySQL

Implemented secure REST APIs, RBAC, parameterized queries, automated profile enrichment, deduplication and searchable alumni records.

YouTube Sentiment & Keyword Analytics | NLP / Backend

Contributed to a real-time analytics platform with multilingual transcript mapping, sentiment classification, keyword tracking and hardened API data handling.

Zero-Shot Hate Speech Detection in Roman Urdu | M.S. Research

Investigated transfer-learning methods for Roman Urdu social-media text analysis, with potential use in OSINT and social media forensics.

Network Traffic Analysis & Threat Detection | DFIR

Analyzed traffic with Wireshark, T-Shark and Snort; mapped indicators of compromise and suspicious communication patterns to investigation findings.

Multilingual OCR & Image Recognition | Computer Vision

Developed document text-extraction workflows using EasyOCR and preprocessing; evaluated pretrained CNNs (VGG16/ResNet) for image recognition tasks.

NLP Toolkit & Predictive Analytics | Applied Machine Learning

Built preprocessing, sentiment and topic-modeling pipelines with NLTK and spaCy; implemented feature engineering, cross-validation and model evaluation using Scikit-learn.

Audio Classification & Emotion Recognition | Deep Learning

Applied MFCC features, CNNs and CNN-LSTM architectures to environmental audio and speech emotion classification; integrated Whisper transcription workflows.

PROFESSIONAL CERTIFICATIONS & TRAINING

- Certified Penetration Testing Professional (CPENT) - EC-Council
- Certified Ethical Hacker (CEH) - NAVTTC / Corvit Rawalpindi
- 5th Digital Pakistan Cybersecurity Hackathon - Ignite, Ministry of IT & Telecom, Trillium | 18-20 Sep 2026
- 5th Digital Pakistan Cybersecurity Hackathon Advanced Workshop - Participant | 15-16 Oct 2026
- OPSWAT Network Security Associate (ONSA) - OPSWAT Academy
- Introduction to Critical Infrastructure Protection (ICIP) - OPSWAT Academy
- Foundation of Threat Hunting - Purple Academy / Picus Security
- MITRE ATT&CK Fundamentals - Purple Academy / Picus Security
- ISO/IEC 27001:2022 Information Security Associate - SkillFront
- Assets, Threats, and Vulnerabilities - Google / Coursera
- Web Designing & Development (6-Month) - Mega IT Solution / PSDF

EDUCATION

M.S. Cyber Security | Air University, Islamabad | 2022–2025

Research focus: Roman Urdu zero-shot transfer learning, digital forensics, network security and threat intelligence.

B.S. Computer Science | Fazaia Bilquis College of Education (Air University affiliated), Rawalpindi | 2017–2021

Capstone: Deep-learning-based image super-resolution.

CYBERSECURITY COMMUNITY & COMPETITIONS

2026 participation: NineByte CTF (Certificate of Achievement), 5th Digital Pakistan Cybersecurity Hackathon, Hackerverse DFIR CTF and BlitzStorm CTF. Active hands-on practice on TryHackMe and Hack The Box.